# 1 2016 1026-ns exam

Vetenskapsteori och vetenskaplig metodik med tillämpningar (beräkningsvetenskap) (Kungliga Tekniska Högskolan)

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# Final exam
## *Theory and Methodology of Science*

Natural science

Course codes
AK2030/AK2036/AK2050/F1N5113/FAK3024

October 26, 2016

## General information

The exam consists of three parts: a 'multiple choice' section (Part I) comprising 20 points, a 'two sentences' section (Part II) comprising 5 points, and a 'short answer' section (Part III) comprising 25 points. Pen/pencil, rubber and paper dictionary are the only allowed external materials.

Grades are awarded according to the following tables:

**Master students**

| **Points** | **Grade** |
|---|---|
| 44.0–50.0 | A |
| 39.0–43.5 | B |
| 34.0–38.5 | C |
| 29.0–33.5 (of which $\geq$ 5 from Part III) | D |
| 25.0–28.5 (of which $\geq$ 5 from Part III) | E |
| 22.0–24.5 (of which $\geq$ 5 from Part III) | FX |
| 0.0–21.5 or <5 from Part III | F |

Bonus points from the preparation and lecture participation questions will be added to the total sum if the exam score is 25 or above.



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**PhD students**

| **Points** | **Grade** |
|---|---|
| 35.0-50.0 (of which $\geq$ 5 from Part III) | Pass (without need for bonus points) |
| 25.0-34.5 (of which $\geq$ 5 from Part III) | Pass if the student's bonus points bring the total sum to 35.0 or above;<br>Extra assignment (FX) required if the student's bonus points bring the total sum to 29.0-34.5;<br>Otherwise Fail |
| 29.0-34.5 (of which $\geq$ 5 from Part III) | Extra assignment (FX) required (if the student's bonus points do not bring the total sum to 35.0 or above) |
| 0.0-24.5 or <5 from Part III | Fail |

The exam is four hours long. You are free to leave after one hour.

Examiner: Prof. Till Grüne-Yanoff, Division of Philosophy, KTH. Visiting address: Brinellvägen 32, top floor.



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# PART I – Multiple choice questions

*Instructions for multiple choice questions:* For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives. Mark your alternatives *clearly* with an 'X'. A box with faint or unclear marks will be interpreted as empty.

1. Does methodology aim to eliminate all errors in science? Select all correct answers.

   a) Yes. Erroneous claims are unscientific.
   b) No. Science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
   c) No. Almost all our beliefs are fallible, so it would be impossible to completely eliminate error.
   d) No. Available evidence often gives us reason to believe, but not always with certainty.

2. Which of the following are necessary conditions for a person X to know proposition P according to the classical definition of knowledge? Select all correct alternatives.

   a) That X is justified in believing P.
   b) That P has sufficient utility for X.
   c) That P is true.
   d) That X believes P with certainty.

3. What is correct about the *hypothetico-deductive* method for hypothesis testing? Select all true claims.

   a) Hypotheses are tested through observable consequences.
   b) The hypothetico-deductive method consists in first testing whether the empirical claim is true and then formulating an hypothesis based on that test.
   c) Operationalization of not directly observable features of interest is necessary for the hypothetico-deductive method.
   d) If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.

4. Which of the following sentences express good hypotheses?

   a) "A square is a geometric shape with three sides."
   b) "Epidemic E was caused by a viral infection."
   c) "Is Amyloid plaque the cause of Alzheimer's?"
   d) "Computer program A is more efficient than program B."



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5. Which of the following claims about random error is correct? Select all true claims.

   a) Random errors can be reduced by repeating measurements and averaging over outcomes.
   b) Random error in an experiment is reduced by better control of background factors.
   c) Random errors are inherently unpredictable, and have null expected value.
   d) Lowering the amount of random errors will increase the amount of systematic errors.

6. Which of the following is true of interval scales? Select all true claims.

   a) It is meaningful to say that a measurement that is two times as large as another measurement represents twice the measured property.
   b) Interval scales are identical up to positive linear transformations
   c) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   d) Time measured in years AD (CE) is an interval scale.

7. Which advantages do experiments have over other empirical methods for testing hypotheses?

   a) Experiments allow to directly test causal hypotheses.
   b) Experiments help exploring phenomena that are hard to observe in nature.
   c) Experiments offer greater control of many relevant background conditions than other empirical methods.
   d) Experiments can be an aid in determining which policy (out of a menu of possible policies) will be most efficient.

8. What is correct about the *repeatability* of experiments? Select all correct alternatives.

   a) The criterion of repeatability is sufficiently satisfied when a person reports that she has done a number of trials with different experimental setups for each trial and out of these trials succeeded in reproducing the same result twice.
   b) Repeatability of an experiment can be determined by analysing the published material about this experiment.
   c) If an experiment is repeatable, then it is also reproducible.
   d) An experiment is repeatable if other competent persons that follow the given description of the experiment can repeat it.



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9. What is the relation between models and experiments? Select all correct claims

   a) Experiments and models are similar in that they always are mere representations of targets.
   b) The problems posed by flawed proofs and programming bugs in models is a greater threat to internal validity than the problem of experimental control poses for experiments.
   c) Models are always representations of targets different from themselves, while experiments are sometimes performed on the target systems themselves.
   d) Internal validity is generally less of a problem for models than for experiments.

10. What distinguishes models from theories? Select all correct claims.

   a) By proposing a theory, one claims that the theory's object is indeed governed by the principles expressed in the theory. When using a model, one is typically aware that alternative representations of the same object useful for other purposes exist.
   b) Scientists who use a theory have reasons to believe that it is not false. Scientists who use a model typically know that it is false in several respects.
   c) In those scientific domains where models and theories co-exist (e.g. chemistry), theories typically express general principles, which are applied in the building of models.
   d) Theories are the source of episteme, models are the source of techne.

11. Which claims are true about Bayesian statistics? Select all correct alternatives.

   a) Bayesian statistics interprets probabilities as "degrees of belief" or "subjective confidence in hypotheses"
   b) The main goal in Bayesian hypothesis evaluation is to decide whether to accept a single hypothesis as either true or false.
   c) In Bayesian statistics, no probabilities are assigned to hypotheses.
   d) In Bayesian statistics, the prior probability of a hypothesis is changed when new evidence is observed

12. Why should hypotheses be tested statistically? Select all correct options.

   a) Statistics is the only way how to justify inductive inferences.
   b) Statistical accounts impose weaker demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.
   c) Neither confirmation nor falsification can be based on deductive relationships of hypotheses and evidence.
   d) Statistics provides tools that turn hypothesis testing into a non-fallible procedure.



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13. Which of the following explanations has the explanans and explanadum in the right order? Select all correct alternatives.

    a) The spherical Earth is rotating on an axis aligned through the poles because water swirls clockwise down a plughole in the northern hemisphere.
    b) The volume of the gas in the container reduced because it was heated.
    c) Water swirls clockwise down a plughole in the northern hemisphere because the spherical earth is rotating on an axis aligned through the poles.
    d) The Earth exerts a gravitational pull on the Moon because the same side of the Moon is always facing the Earth.

14. Which claims are true about the relation between correlation and causation? Select all correct claims.

    a) Mill's method of difference has been largely ignored by the modern practice of randomised controlled trials.
    b) Correlation measures the covariance between variables. Causation measures the influence of one variable on another.
    c) By analysing correlation data, we typically are able to determine which causal model is the unique explanation of the data.
    d) Variables can be correlated with each other even though no causal relation exists between them.

15. What are the main claims of anti-naturalist views of the social sciences? Select all correct alternatives.

    a) The social sciences must employ their own methods, different from the natural sciences - but both the natural and the social sciences can use the same methodology.
    b) The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalise from them.
    c) Although the social sciences use qualitative data, in contrast to the natural sciences, there are methods by which this data can be quantified and thus be analyzed with natural science methods.
    d) The social sciences cannot and should not follow the methodology of the natural sciences.

16. What is true about belief-desire explanations? Select all correct alternatives. Belief-desire explanations ...

    a) ... can be used to explain any system's or organism's actions.
    b) ... are often used to explain human behavior.
    c) ... entail that the ends always justify the means.
    d) ... are less precise than causal explanations.

17. What is true about *scientific openness*? Select all correct alternatives. Not sharing scientific data may be justified when ...

    a) ... protecting the institution's moral credibility.
    b) ... the data reveals scientific misconduct.
    c) ... protecting intellectual property.
    d) ... additional data checks are necessary.



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18. What is true about the *precautionary principle*? Select all correct answers. The precautionary principle ...

    a) ... states that when laboratorial safety regulations cannot be fully met, experiments should not be conducted.
    b) ... states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically.
    c) ... states a list of research projects that should never be initiated.
    d) ... imposes a moral constraint on researchers.

19. In the risk discourse, which are the *quantitative* interpretations of risk? Select all correct alternatives.

    a) The severity of an unwanted event.
    b) The cause of an unwanted event.
    c) The probability of an unwanted event which may or may not occur.
    d) The statistical mean value of possible harmful outcomes.

20. What is true about how risks are perceived? Select all correct alternatives. Psychological studies have shown that ...

    a) ... risk to identifiable individuals are seen as worse than statistical risks (with same or greater number of expected deaths).
    b) ... risks where one lacks control are seen as worse.
    c) ... involuntary risks are seen as worse than voluntary risks.
    d) ... natural risks are seen as worse than technological risks.



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## PART II – Two-sentences questions

*Instructions for two-sentences questions:* Write one or two sentences for each of the questions below on a separate sheet of paper, *not* on the exam. One point will be awarded for each correct answer. A partially correct answer will be awarded half a point. An incorrect answer, or an omitted answer, gives no points. Writing more than two sentences is not needed, but will not lead to a deduction of points. However, an answer that has both correct and incorrect parts indicates that you have not fully understood the right answer and will result in no points awarded.

1. What are the three criteria of informed consent?
2. What is a false-positive (Type I) error?
3. Name one thing that is only observable indirectly and one thing that is observable directly.
4. What is a double-blind test, in a medical context?
5. What is consequentialism?



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## PART III – Short answer questions

The following questions should be answered on separate sheets of paper. *Hints on answer lengths*: If you answer a question worth 5 points with a single sentence, it is unlikely that your answer is elaborate enough to be awarded 5 points. On the other hand, if your answer is informative and to the point, there should be little use for more than one page for a 5 points question and two pages for a 10 points question.

1. Discuss the claim: "If a theory makes a prediction that contradicts measured data then the theory should be rejected". Is this claim true of actual scientific practice? Should the claim be true of good scientific practice? (5 p)

2. Frederick Banting and Charles Best are often credited with the discovery of the connection between diabetes and insulin.

   (a) Before their experiments, insulin had never been obtained and remained a hypothetical substance. Banting believed that earlier failures to detect insulin were due to the fact that the insulin was destroyed by a substance called tryptine. Was this an *ad-hoc hypothesis* or not? Explain and motivate your answer. (1 p)

   (b) Banting and Best tried to determine if the test animals had diabetes by measuring and calculating the ratio between glucose and nitrogen in the urine. What is it called when one tries to make a concept measurable? What are the conditions for doing so in a good way? (2 p)

   (c) Experimenting on dogs instead of humans was done because of practical and ethical reasons. This affected the two different kinds of conclusions they could draw from the experiment. What advantages and disadvantages would such a decision have, and how could you frame this in methodological terms? (2 p)

3. Consider the following definition: "X is a traffic accident if and only if X is an accident between two vehicles in traffic." Assuming that we treat this as a *lexical* definition, what problems do you see with the definition? What would the difference be if we treated it as a *stipulative* definition? Would there still be problems with the definition? (5 p)

4. A dowser claims to be able to find concealed metallic objects in fields by forming an inner picture of the object and walking around with his dowsing rod. When he gets close enough to the object (less than 1 m), he is bound to find it, he claims. You are to design an experiment for determining whether the dowser really can find metallic objects with the dowsing rod.

   (a) State the hypothesis you wish to test and why it is good. Also state at least one auxiliary hypotheses and why you need it. (1.5 p)

   (b) List and explain those methods you think are necessary to use in order to ensure high internal validity when testing this hypothesis. (4.5 p)



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(c) Explain what it would take for an experiment testing the hypothesis to be reproducible. (1 p)

(d) What is necessary for a statistical evaluation of the result of an experiment testing the hypothesis? Explain and argue for your answer. (You are not expected to account for the steps of statistical hypothesis testing nor make any calculations in your answer.) (3 p)

Please note that your resources are near unlimited when you are testing this claim and that you should always use the best possible methods. Also note that this is a 10 points question, so we expect a rather careful answer in terms of argumentation.



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# TaMoS: Answer sheet for mcq's, Natural science

Student's name:

Personal number:

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| | a) | b) | c) | d) |
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